INCH-POUND

MIL-A-70945(AR)

30 July 1990

MILITARY SPECIFICATION

ANTENNA ASSEMBLY: 12922146

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Antenna Assembly, used in the M90 Radar Chronograph (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for

MIL-P-14232 - Parts, Equipment and Tools for Army Materiel, Packaging of

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1290

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-2164 - Environmental Stress Screening Process for Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC).

12922098 - Bracket Assembly
12922105 - EMI Test Procedure
12922126 - Microstrip Antenna
12922146 - Antenna Assembly
12922225 - Environmental Test Methods

SPECIAL PACKAGING INSTRUCTIONS

P12922146 - Special Packaging Instructions for the Antenna Assembly

(Copies of drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General description. The antenna is an input/output device that transmits an RF signal upon command from the M90 Radar Chronograph data unit and receives the doppler signal reflected from the projectile providing signal to the data unit for processing.

3.2 Materials. The materials shall be in accordance with the applicable drawings and specifications.

3.3 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12922146, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Antenna Assembly shall meet the requirements of MIL-F-13926 and MIL-STD-454 as follows :

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship
e. Soldering electrical connections

3.4 Performance.

3.4.1 General performance.

3.4.1.1 Input power. Input power shall be supplied to the Antenna Assembly on pins A and B of the connector J1. The Antenna Assembly shall withstand the following limits:

a.	Minimum voltage:	24 VDC.
b.	Maximum voltage:	55 VDC.
c.	Consumption:	18 Watt maximum.

3.4.1.2 ON/OFF control. The ON/OFF control command to the Antenna Assembly shall be supplied on pins D and E of input connector J1. The command shall be a discrete 12 VDC +/-0.6 V input, with 680 ohm series resistors on command and return lines.

a. "ON" command: Antenna transmit and receive.
b. "OFF" command: Antenna shutdown.

3.4.1.3 Doppler output signal. Two, AC coupled, 180 degrees relative phase difference, Doppler signal outputs shall be provided. The outputs shall appear on pin D (DOPPLER +) and pin E (DOPPLER -). The signal shall satisfy the following conditions on both outputs:

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a. Noise: 2 VPTP maximum.
b. Saturated Output Swing: 14 VPTP minimum, 18 VPTP maximum.
c. Bandwidth: 7 KHz to 100 KHz.

3.4.1.4 Common line interface. The ON/OFF control input signal and the DOPPLER signal output (+ and -) shall be common lines. Figure 1 shall be used as a reference for fabrication of the common lines interface.

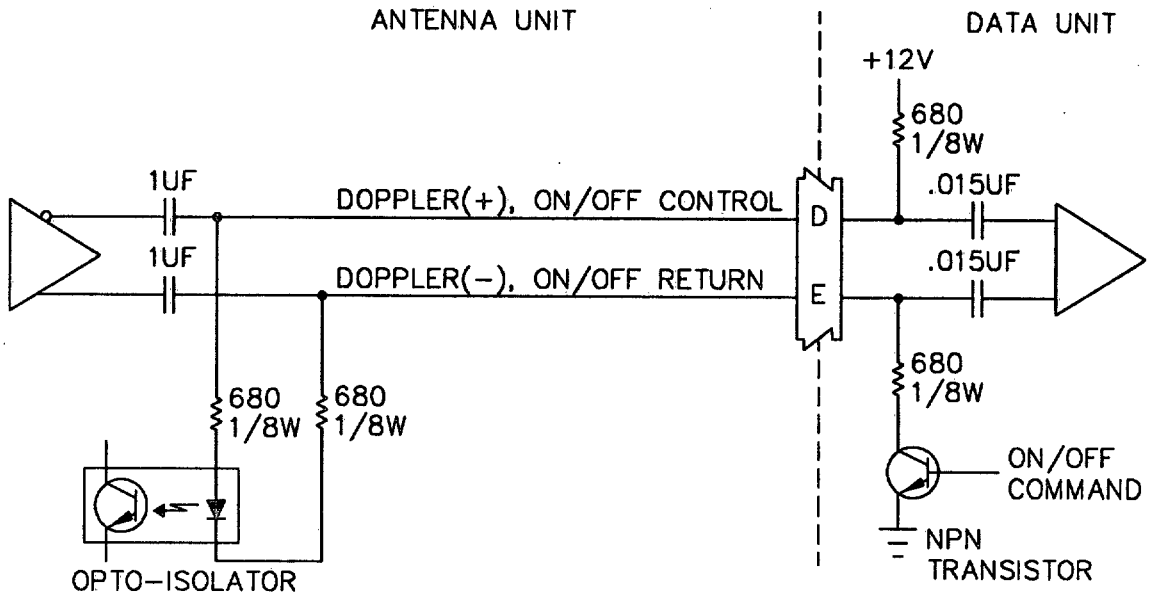


FIGURE 1 - COMMON LINES INTERFACE

3.4.2 Microwave radiating pattern performance. The Antenna Assembly shall satisfy the following requirements for microwave radiating pattern, using a microstrip antenna in accordance with Drawing 12922126.

a. Gain: 22 DB minimum.
b. -3 DB beamwidth, E-Plane-horizontal: 15 +/- 2 degrees.
c. -3 DB beamwidth, H-Plane-vertical: 7 +/- 2 degrees.
d. Side lobe level: -12 DB maximum below main lobe.
e. Squint angle, E-Plane: 0 +/-1 degrees.
f. Squint angle, H-Plane: 7 +/-1 degrees.
g. Cross polarization: -24 DB maximum.

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3.4.3 Receive/transmit. The Antenna Assembly shall satisfy the following Receiving/Transmitting requirements:

a. Frequency:
Frequency shall be factory tuned to 10525 +/- 1 MHz at 77 +/- 14 degree F Ambient.
Frequency drift vs Time: +/- 5MHz maximum per 6 month period.
Frequency drift vs Temp: -50 degree F to 125 degree F: +/- 7 MHz maximum.

b. Output power: 20.8 DBM minimum to 23.8 DBM maximum under all environmental conditions, at the microstrip antenna connector.

c. Receiver sensitivity: -106 DBM minimum.

d. Receiver bandwidth: 7 KHz to 100 KHz.

3.4.4 Input/output interconnections. Input/output connections to the Antenna Assembly shall be with a 5 pin connector type MS3124E-14-5PY. Pin assignment shall be in accordance with TABLE I.

TABLE I. Wiring connections

PIN	SIGNAL
A	24-55VDC
B	GND
C	SHIELD
D	DOPPLER(+), ON/OFF RETURN
E	DOPPLER(-), ON/OFF CONTROL

3.5 Environmental requirements.

3.5.1 High temperature. The Antenna Assembly shall meet the requirements of 3.3 and 3.4 when thermally stabilized at +125 degrees F. The Antenna Assembly when subjected to storage temperature of +150 degrees F shall meet the requirements of 3.3 and 3.4 when returned and thermally stabilized at ambient temperature.

3.5.2 Low temperature. The Antenna Assembly shall meet the requirements of 3.3 and 3.4 when stabilized at -50 degrees F and when returned and thermally stabilized at ambient temperature.

3.5.3 Solar Radiation. The Antenna Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to 355 BTU per square foot.

3.5.4 Humidity. The Antenna Assembly shall meet the requirements of 3.3 and 3.4 after being subject to 240 hours at a

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relative humidity of 94 +/-5 percent from +68 degrees F to +125 degrees F.

3.5.5 Shock. The Antenna Assembly when mounted on Bracket P/N 12922098, shall meet the requirements of 3.3 and 3.4 after being subjected to a shock of 50 G's for 11 milliseconds duration in each of its mutually perpendicular axis for a total of 18 shocks.

3.5.6 Vibration. The Antenna Assembly when mounted on Bracket P/N 12922098 shall meet the requirements of 3.3 and 3.4 after being subjected to vibration, having a double amplitude of 1.0 inch, from 5 to 6.5 HZ and acceleration level of 2 G's over the 6.5 to 200 HZ frequency range for a 15-minute duration in each of its three mutually perpendicular axes.

3.5.7 Altitude. The Antenna Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to and altitude of 10,000 feet in operational mode and after exposure at an altitude of 50,000 feet in nonoperational mode.

3.5.8 Rain. The Antenna Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to a 2.0 inch per hour, 20 mile per hour wind driven rain.

3.5.9 Salt fog. The Antenna Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to 48 hours of continuous exposure to a 5 percent salt solution with a specific gravity of 1.027 to 1.034, a PH of 6.5 to 7.0 and a temperature of 95 degrees F.

3.5.10 Sand and dust. The Antenna Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to 36 hours of exposure to wind driven sand and dust.

3.5.11 Fungus. The Antenna Assembly shall meet the requirements of 3.3 and 3.4 after being subjected to 28-day fungus exposure to a composite spore suspension.

3.5.12 EMI. The Antenna Unit shall meet the requirement of drawing 12922105 when connected to an M90 RADAR CHRONOGRAPH (NSN: 1290-01-073-0764), supplied as Government-furnished material.

3.6 Environmental stress screening (ESS). The Antenna Assembly shall be subjected to a 40-hour run-in period (excluding downtime) which includes vibration and operation at temperature extremes. The time required to conduct the vibration test shall not be included as part of-the-run-in period. The Antenna

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Assembly shall meet the requirements of this specification for the 40-hour run-in period.

3.6.1 Environmental stress screening (ESS) temperature. The Antenna Assembly shall meet the performance requirements in 3.3 and 3.4 after being subjected to 40-hours of temperature cycling between 0 degrees F and +125 degrees F in accordance with 5.1.2.1 of MIL-STD-2164.

3.6.2 Environmental stress screening (ESS) vibration. The Antenna Assembly shall meet the performance requirements in 3.3 and 3.4 after being subjected to random motion in the range of 20 to 2000 Hz for 5 minutes at an axis perpendicular to the largest surfaces of the Antenna Assembly. Acceleration spectral density is 6.06 g. RMS in accordance with 5.1.1 of MIL-STD-2164.

3.7 Fabrication. The Antenna shall be manufactured in accordance with Drawing 12922146.

3.8 First article inspection. When specified in the contract or purchase order, three samples shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.9 Workmanship. The workmanship requirements of MIL-F-13926 and MIL-STD-454 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 and all applicable drawings. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all

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requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 and MIL-STD-454 form a part of the Quality Assurance Provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60 degrees to +90 degrees F.
b.	Relative humidity	90% RH maximum.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three Antenna Assemblies for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in TABLE II and TABLE III herein, and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any Antenna Assembly to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements.

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Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in TABLE II shall be performed as noted under the characteristics.

TABLE II. Functional tests
(100% Inspection)

No.	Characteristics	Requirement	Test Procedure
Critical:	None	None	N/A
Major:			
101.	Performance	3.4	4.8
102.	ESS Vibration	3.6.1	4.7.1.1
103.	ESS Temperature	3.6.2	4.7.1.2

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No.	Characteristics	Requirement	Test Procedure
104.	Fabrication	3.7	Applicable Drawing/ Visual
105.	Materials	3.2	Applicable Drawing/ Visual
106.	Workmanship	3.9	Applicable Drawing/ Visual

Minor: None

4.5 Special sampling. Three Antenna Assemblies shall be selected at random as a special sample from each lot or each 150 produced whichever occurs first. The sample shall meet the requirements and tests of TABLE III.

TABLE III. Environmental sampling tests.

No.	Characteristics	Requirement	Test Procedure
301.	High temperature	3.5.1	4.7.2.1
302.	Low temperature	3.5.2	4.7.2.2
303.	Solar Radiation	3.5.3	4.7.2.3
304.	Humidity	3.5.4	4.7.2.4
305.	Shock	3.5.5	4.7.2.5
306.	Vibration	3.5.6	4.7.2.6
307.	Altitude	3.5.7	4.7.2.7
308.	Rain	3.5.8	4.7.2.8
309.	Salt Fog	3.5.9	4.7.2.9
310.	Sand and Dust	3.5.10	4.7.2.10
311.	Fungus	3.5.11	4.7.2.11
312.	EMI	3.5.12	4.7.2.12

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.4.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the

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component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.4.

4.6.2.3 Inspection equipment design requirements.

4.6.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5 degrees F (1.4 degrees C). The temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5 degrees F per foot in any direction. The heat source of the test chamber shall be located so that radiant heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber within the supply airstream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.6.2.3.2 Fabrication, setup and calibration. Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instructions shall be prepared and submitted to the procuring activity.

4.7 Methods of inspection. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the government procuring agency for approval prior to use.

4.7.1 Environmental stress screening.

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4.7.1.1 ESS temperature. Testing equipment utilized in this test shall conform to the requirements of 4.6.2.3.1. The Antenna Assembly shall be placed in the chamber and subjected to a 40-hour run-in period which includes vibration and operation at temperature extremes. The temperature shall be raised gradually at a rate not to exceed 2 degrees F per minute until thermally stabilized at +125 degrees F for 4 hours duration. Then the temperature of the chamber shall be decreased to 0 degrees F at the same rate for another 4 hours. Subsequent to the above temperature extremes (+125 degrees F and 0 degrees F), the Antenna Assembly shall not have been damaged or adversely affected and shall meet the requirements in 3.3 and 3.4 of this specification for the entire 40-hour run-in period.

4.7.1.2 ESS vibration. The Antenna Assembly in an operating mode shall be exposed to a 5 minute burst of random vibration. Failures occurring during this five minute test shall be accrued (if possible) and corrected at the conclusion of the 5 minute period in accordance with the requirements of 3.3 and 3.4. The random vibration spectrum shall be:

20 - 80 Hz	at 3 dB/octave rise
80 - 350 Hz	at 0.04 g sq/Hz
350-2000 Hz	at 3 dB/octave rolloff

The motion shall be applied perpendicular to the largest surfaces of the Antenna Assembly.

4.7.2 Environmental methods of inspection.

4.7.2.1 High temperature. This test shall be conducted in accordance with Drawing 12922225. Subsequent to the exposure, the Antenna Assembly shall not be damaged or adversely affected and shall meet the requirements in 3.5.1 of this specification.

4.7.2.2 Low temperature. This test shall be conducted in accordance with Drawing 12922225. Subsequent to the exposure, the Antenna Assembly shall not be damaged or adversely affected and shall meet the requirements in 3.5.2.

4.7.2.3 Solar radiation. This test shall be conducted in accordance with Drawing 12922225. Subsequent to the exposure, the Antenna Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.3 of this specification.

4.7.2.4 Humidity. The Antenna Assembly shall be subjected to the humidity test as outlined in Drawing 12922225. Performance test to be conducted during the exposure shall be

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conducted during the first, third and fifth cycles; subsequently, the Antenna Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.4 of this specification.

4.7.2.5 Shock. The Antenna Assembly shall be subjected to three 50g, 11 millisecond duration, 1/2 sine shock pulse in each of the three mutually perpendicular directions for a total of 18 shocks as outlined in Drawing 12922225. The Antenna Assembly shall be energized throughout the test. Subsequent to the shock test, the Antenna Assembly shall not have been damaged or adversely affected and shall meet the performance requirements in 3.5.5 of this specification.

4.7.2.6 Vibration. The Antenna Assembly shall be subjected to the vibration as specified in Drawing 12922225. The displacement and frequency shall be as follows:

GRAVITY	FREQUENCY (Hz)
1.0 inch Double amplitude	5 - 6.5
2G Acceleration	6.5 - 200

Subsequent to the vibration test, the Front Panel Assembly shall not have been damaged or adversely affected and shall meet the performance requirements in 3.5.6 of this specification. Duration shall be one sweep of 15 minutes in each plane.

4.7.2.7 Altitude. The Antenna Assembly shall be subjected to altitude test as specified in Drawing 12922225. Subsequent to the exposure, the Antenna Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.7 of this specification.

4.7.2.8 Rain. The Antenna Assembly shall be subjected to rain test as specified in Drawing 12922225. Subsequent to the exposure, the Antenna Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.8 of this specification.

4.7.2.9 Salt fog. The Antenna Assembly shall be subjected to the salt test as specified in Drawing 12922225. Subsequent to the exposure, the Antenna Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.9 of this specification.

4.7.2.10 Sand and dust. The Antenna Assembly shall be subjected to the sand and dust test as specified in Drawing

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12922225. Subsequent to the exposure, the Antenna Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.10 of this specification.

4.7.2.11 Fungus. The Antenna Assembly shall be subjected to the fungus test as specified in Drawing 12922225, for 28 days. Subsequent to the exposure, the Antenna Assembly shall not be damaged or adversely affected and shall meet the performance requirements in 3.5.11 of this specification.

4.7.2.12 EMI. The Antenna Assembly shall be tested in accordance with test procedure of Drawing 12922105.

4.8 Performance. The performance parameters of 3.4.1 through 3.4.4 inclusive, shall be demonstrated using contractor designed, Government-approved, test fixture simulating the conditions and making the measurements. The performance test shall verify that each of the following Antenna Assembly functional characteristics conform to the requirements of 3.4.1 through 3.4.4:

4.8.1 Input power. The Input Power shall function satisfactorily with input voltages in accordance with 3.4.1.1 (a,b).

4.8.2 Consumption. Power consumption shall not be greater than requirement of 3.4.1.1 (c).

4.8.3 ON/OFF control. The ON/OFF Control command shall transmit RF-energy upon receipt of "ON" command, in accordance with 3.4.1.2.

4.8.4 Transmitted frequency. Transmitted frequency shall be in accordance with 3.4.3 (a).

4.8.5 Transmitted power. Transmitted power shall be in accordance with 3.4.3 (b).

4.8.6 Receiver sensitivity. Receiver sensitivity shall be in accordance with 3.4.3 (c,d).

4.8.7 Doppler output signal. Doppler output signal shall be in accordance with 3.4.1.3.

4.8.8 Microwave radiating pattern performance. The Antenna Assembly shall be tested in accordance with 3.4.2 and in an antenna range at the microstrip antenna connector (DWG 12922126). Production antenna shall be tested at 10525 MHz. First Article Test and special sampling antenna shall be tested at 10510, 10525 and 10540 MHz.

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5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing and marking shall be in accordance with MIL-P-14232 and Special Packaging Instruction P12922146. The level of protection shall be specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Antenna Assembly is intended for use in the M90 Radar Chronograph.

6.2 Acquisition requirements. Procurement data should specify the following :

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Applicable stock number.
d. First Article Sample requirements (see 3.8).
e. Level of preservation and packing (see 5.1).
f. Serialization requirements, if applicable.
g. Certificate of conformance for each lot or shipment of product.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirement of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DOD Form 1423.

Reference Paragraph	DID Title	DID No.	Suggested Tailoring
4.6	Quality Engineering Acceptance Inspection Equipment Description Documentation	DI-R-1714	None

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 or as stated in the contract will apply.

6.6 Definitions.

6.6.1 Contractor. Unless otherwise specified, the word Contractor used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government Owned-Contractor Operated (GOCO), and Government Owned-Government Operated (GOGO) plants.

6.6.2 Technical data package. A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable lists, specifications, standards, models, performance requirements, quality assurance provisions and packaging data.

6.7 Government furnished property. The contracting officer should arrange to furnish the property listed in 3.4.5.

6.8 Subject term (key word) listing.

M90 Radar Chronograph (M90)
Antenna Assembly (Antenna)

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1290-A457)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-A-70945 (AR)	30 July 1990

3. DOCUMENT TITLE
ANTENNA ASSEMBLY: 12922146

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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